

Economic decision analysis for genomic medicine

A frozen public-genome benchmark demonstrating how GenomeLens translates genomic evidence into decision-relevant health economics while preserving uncertainty, refusal states, and auditability.

36 / 36

PUBLIC GENOMES COMPLETED

7,723

TESTS PASSED

0

ARITHMETIC DISCREPANCIES

301

REAL-ROW ASSERTIONS | 0 FAILURES

144

RENDERED PDFs CHECKED

Executive summary

Validation layer	Frozen v1 result	Interpretation
Cohort completion	36/36	All intended public benchmark genomes completed on pinned trees.
Decision routing	9/9 gates; 0 parity failures	Structured outputs and rendered economic statements reconcile.
Zero semantics	0 defects across 1,399 displayed zeros	Zero is reserved for supported null/no-change states, not missing information.
Economic arithmetic	0 discrepancies	All recomputable economic values agreed with their recorded components.
Report integrity	36/36 PDF checks; 0 forbidden hits across 144 PDFs	Public-facing economic reports passed semantic and disclosure checks.

Core result. GenomeLens v1 completed a frozen, reproducible public-data benchmark with no known P0/P1 defect in the economic decision layer. The benchmark is a technical proof of concept, not a claim of diagnostic accuracy, patient benefit, or regulatory validation.

What is being demonstrated

GenomeLens is designed to answer a question that sits between genomic interpretation and conventional health-economic evaluation: **if a genomic finding is credible, what decision could it change, what health and cost consequences could follow, and how should uncertainty be represented?** The benchmark tests that translation layer rather than claiming a single monetary value for a genome.

GENOMELENS | CAPABILITY FRAMEWORK

From genomic evidence to economic decision support

The public benchmark is organized around explicit decision states. It demonstrates the outputs GenomeLens can produce without disclosing proprietary routing logic, internal parameter registries, or implementation details.

1. Genomic evidence	2. Decision context	3. Health consequence	4. Economic consequence	5. Uncertainty
Evidence-qualified finding	Intervention / comparator	Outcome change	Δ Cost, Δ QALY, ICER, NMB	Sensitivity / VOI / unresolved

Two public-facing analysis modes

Mode	Unit of analysis	Decision outputs	Economic outputs
Individual Genomic Economics	One public genome	Screening, surveillance, treatment selection, pharmacogenomic or other context-dependent decisions	Incremental cost, QALY impact, ICER/NMB where supported; conditional or unquantified states where not
Reproductive & Carrier Economics	Two biologically paired parental genomes	Inheritance compatibility, offspring risk, targeted testing, prenatal/PGT-M or other reproductive decision pathways	Expected cost/outcome consequences for the pair-level decision; kept separate from individual-health value

What makes the decision layer auditable

Design principle	Public-facing behavior
Evidence state is separate from economic state	A finding can be supported biologically but still conditional, unquantified, or refused economically.
Missing is not zero	Technically unresolved or context-missing pathways are not silently displayed as \$0.
Estimands stay separate	Present value, if/when-decision value, reproductive value, family/cascade value, and scenario values are not silently summed.
No-change states remain visible	Guideline-based non-action or measured nulls can appear as genuine zeros with an explicit reason.
Public by construction	The benchmark exposes outputs and validation evidence, not proprietary routing logic or private input data.

IP boundary. This brief intentionally documents capabilities, validation results, estimands, and output semantics while omitting proprietary implementation details needed to reproduce the engine.

GENOMELENS | BENCHMARK DESIGN

A public-data benchmark built to test the decision layer

The frozen benchmark contains 30 primary public genomes plus 6 reference/parent genomes. Where donor context is not observed, public scenarios are standardized rather than presented as personalized patient estimates.

Layer	Question	Frozen evidence
1. Provenance	Did the benchmark run the intended public inputs at the intended code state?	36/36 samples pinned to engine 4eaf729 on clean worktrees; input hashes verified against the benchmark baseline.
2. Routing	Did findings enter the correct modeled / conditional / unresolved / refused state?	9/9 release gates; 0 routing/parity failures.
3. Economics	Do cost/QALY/NMB calculations reconcile where source terms permit recomputation?	88 values recomputed with 0 discrepancies; 20 explicitly non-recomputable source-derived null/context rows.
4. Reporting	Do PDFs communicate the same values and states as the structured result?	36/36 PDF checks; 301/301 real-row assertions; 0 forbidden-string hits across 144 rendered PDFs.
5. External challenge	Can an independent language-model review inject unsupported results into the economics?	278/278 analyst items answered; all 10 synthetic adversarial cases rejected; no payload changed.

Release evidence at a glance

Metric	Result
Full clean-tree suite	7,723 passed 26 skipped 0 failed
Zero audit	0 defects across 1,399 displayed zeros
Arithmetic audit	0 discrepancies
Lost-content audit	0 downgraded rows
Real-row assertions	301 evaluated 0 failures
Analyst review	278/278 schema-clean no P0/P1 finding
Economics PDF disclosure scan	0 hits across 144 PDFs

Visual-gate disposition. 49 of 9,482 rendered pages were flagged in the full-genome report for a bounding-box false positive and minor ancestry-label overlap/clipping. The economics PDFs were clean. The issue was accepted as a non-material P2 presentation exception, not recorded as an unconditional visual pass.

GENOMELENS | OUTPUT MODEL

What a GenomeLens economic result contains

The benchmark demonstrates a structured result model rather than a single headline dollar figure. A public result can be positive, negative, zero, conditional, unresolved, or unquantified depending on the evidence and decision context.

Result element	What a reviewer can inspect	Why it matters
Finding + evidence state	Evidence-qualified genomic finding and qualification status	Separates biological support from downstream economic interpretation.
Decision / comparator	The action that could change and the relevant alternative	Economic value is attached to a decision, not to genotype alone.
Health effect	QALY or other modeled health consequence where supportable	Makes the clinical consequence explicit.
Cost effect	Incremental healthcare cost or source-derived economic term	Prevents “value” from being conflated with cash savings.
Economic result	NMB, ICER, or explicitly non-recomputable source-derived value	Supports cost-effectiveness interpretation without forcing every pathway into one metric.
Uncertainty / applicability	Age, sex, drug exposure, technical callability, source limitations, or scenario dependence	Shows when a result applies and when it should not be monetized.

Family-planning economics

The reproductive workflow treats the **parent pair** as the unit of analysis. It evaluates whether parental variants can produce the same clinically meaningful phenotype in offspring, derives the relevant inheritance probability, then maps that risk to reproductive decisions and expected economic consequences. Pair-level value is not added to either parent’s personal-health NMB.

Parent A + Parent B	Inheritance compatibility	Offspring probability	Decision option	Expected economic consequence
Exact public variants / callability	Mechanism + phenotype match	Genotype and phenotype risk	Resolve uncertainty / test / PGT-M / prenatal / no change	Expected cost, health effect, NMB or explicit non-quantification

A deliberate refusal can be a successful result

Example behavior. If a pathway lacks the donor context, intervention evidence, separable health terms, or technical resolution needed for a defensible estimate, GenomeLens can preserve the finding while refusing to assign a headline economic value. That behavior is part of the benchmark, not a failure mode.

GENOMELENS | INTERPRETATION

What Benchmark v1 establishes - and what it does not

The benchmark establishes	The benchmark does not establish
Reproducible execution across 36 intended public genomes	Population prevalence or representativeness
Consistent routing between finding states and economic dispositions	Clinical efficacy of an intervention in a specific patient
Arithmetic reconciliation for recomputable economic results	Diagnostic sensitivity/specificity of whole-genome sequencing
Explicit handling of true zero, missing context, uncertainty, and refusal	A single "cash value" of a genome
Report/structured-result consistency across public economics PDFs	Regulatory clearance or medical-device validation
Pair-level reproductive decision analysis as a distinct estimand	Personalized reproductive advice for real individuals

Disclosed release limitations

Limitation	Disposition
Mutation harnesses were not rerun because source anchors were stale	Recorded as future validation work; not represented as completed.
No separate full manual adversarial reviewer pass of every final artifact	One Lynch report was read end-to-end; automated and analyst checks remain documented separately.
Minor full-report visual defects	Accepted non-material P2 exception; economics PDFs clean.
20 economic rows not independently recomputable from separable $\Delta QALY$ / $\Delta Cost$ terms	Kept explicitly classified rather than back-filled with fabricated terms.

Interpretation standard. Benchmark v1 should be described as a frozen public-data technical proof of concept for genomic health-economic decision analysis. It should not be described as clinical validation, diagnostic performance validation, population effectiveness, or regulatory evidence.

GENOMELENS | REPRODUCIBILITY & PUBLICATION

Frozen release record

Release field	Frozen v1 record
Benchmark	GenomeLens Deterministic Benchmark v1
Engine commit	4eaf729
Tooling commit	a9b3d3a
Cohort	30 primary public genomes + 6 reference/parent genomes
Archive note	Frozen benchmark artifacts were archived and integrity-checked locally.
	Internal benchmark packages and raw run outputs are not publicly distributed.
Personal/private inputs	Excluded from the public benchmark artifact

Public benchmark summary

GenomeLens Deterministic Benchmark v1 demonstrates a reproducible decision layer that connects public whole-genome findings to evidence-qualified clinical decisions and health-economic outcomes. Across 36 public benchmark genomes, the frozen release completed with 0 routing/parity failures, 0 unexplained-zero defects, 0 arithmetic discrepancies among recomputable values, and 301/301 current-contract assertions. The benchmark also demonstrates explicit refusal and uncertainty states, pair-level reproductive economics, and report-to-structured-output consistency.

Publication scope. This public benchmark presents validated outputs, methods, limitations, and reproducibility evidence while excluding proprietary implementation details, private model logic, internal routing tables, private genomic inputs, personal bloodwork, credentials, and other non-public materials.

Release status

DETERMINISTIC BENCHMARK v1 - FROZEN

Public-data technical proof of concept. Prepared for a public repository update after review of the accompanying public materials.